Quantum Field Theory

Lecture Notes

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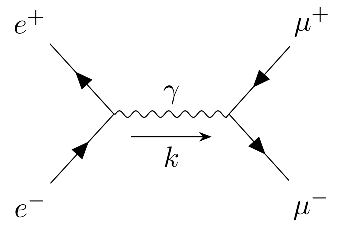

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Lecture 32: Feynman diagrams in Momentum space

Recall that the Feynman propagator and its Fourier transform were defined as

$$D_{\mathrm{F}}(x-y) = \int \frac{\mathrm{d}^4k}{(2\pi)^4}\, \frac{i}{k^2-m^2+i\epsilon} e^{-ik\cdot(x-y)} \qquad \widetilde{D}_{\mathrm{F}}(k) = \frac{i}{k^2-m^2+i\epsilon}$$

Then, if we convert the correlation function to the momentum space, we would have product of these propagators and also, we would introduce a delta function for any internal vertex. For example,

$$\begin{aligned}\int \mathrm{d}^4z\, \triangle(x-z)\triangle(y-z) &\sim \int \mathrm{d}^4z\, \frac{\mathrm{d}^4p_1}{(2\pi)^4}\frac{\mathrm{d}^4p_2}{(2\pi)^4}\widetilde{D}_{\mathrm{F}}(p_1)\widetilde{D}_{\mathrm{F}}(p_2)e^{-ip_1\cdot x - ip_2\cdot y}e^{-i(p_1+p_2)\cdot z} \\ &\sim \int \frac{\mathrm{d}^4p_1}{(2\pi)^4}\frac{\mathrm{d}^4p_2}{(2\pi)^4}\widetilde{D}_{\mathrm{F}}(p_1)\widetilde{D}_{\mathrm{F}}(p_2)\delta^{(4)}(p_1+p_2)e^{-ip_1\cdot x - ip_2\cdot y}\end{aligned}$$

Note that the delta function in some sense, denotes *momentum conservation*. The Feynman rules for the correlation function in momentum space are as follows:

- For each internal line, we associate a momentum k and the propagator $\widetilde{D}_{\mathrm{F}}(k)$
- For each vertex, we write down the factor $(-i\lambda)(2\pi)^4\delta^{(4)}\left(\sum_i k_i\right)$ where $\sum k_i$ is the sum of all momenta flowing *into* the vertex that we are considering (*momentum conservation*).
- Divide by the symmetry factor and the other things that we did before.

The best thing would be to demonstrate through an example. Consider the diagram,

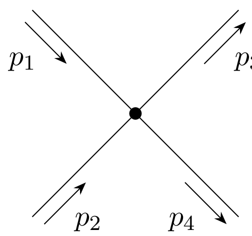


Figure 1: Diagram of order $\mathcal{O}(\lambda)$ for the four-point correlation function with interaction.

We can put the arrows of the momentum in any direction for now, the second rule will incorporate the sign automatically. The contribution of this diagram to the four-point correlation function, according to the above rules would be,

$$(-i\lambda)(2\pi)^4\left(\prod_{j=1}^{4}\frac{i}{p_j^2-m^2+i\epsilon}\right)\delta^{(4)}(p_1+p_2-p_3-p_4)$$

Now, consider a bit more involved diagram, containing a loop.

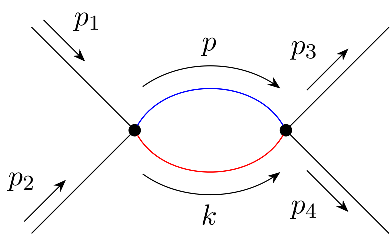


Figure 2: Diagram of order $\mathcal{O}(\lambda^2)$, containing a loop, for the four-point correlation function with interaction.

Let us first impose the momentum conservation, which will give us $\delta^{(4)}(p_1+p_2-p-k)$ and $\delta^{(4)}(p+k-p_3-p_4)$. Note that, these delta functions will be in product with each other. The argument of the second delta function becomes zero for $p=p_3+p_4-k$ and we can substitute this in the first delta function, leading to $\delta^{(4)}(p_1+p_2-p_3-p_4)$.

We see that, both p and k cannot be uniquely determined just from momentum conservation and hence, there is always a dangling momentum remaining, which we need to integrate out.

Also, we can interchange the red and blue lines, leaving the diagram unchanged, giving us a symmetry factor of 2. Thus, considering all the facts, the contribution of the diagram becomes:

$$\frac{1}{2}(-i\lambda)^2(2\pi)^4\delta^{(4)}(p_1+p_2-p_3-p_4)\int\frac{\mathrm{d}^4k}{(2\pi)^4}\frac{i}{k^2-m^2+i\epsilon}\frac{i}{(p_3+p_4-k)^2-m^2+i\epsilon}\left(\prod_{j=1}^{4}\frac{i}{p_j^2-m^2+i\epsilon}\right)$$

1.1 Scattering Matrix

We now focus on how QFT can be used to extract information from physical processes, particularly *scattering*. Consider a state $|i\rangle$ which is at $t\to-\infty$, describing particles, which evolves in time and goes to a final state $|f\rangle$ at $t\to+\infty$.

In the initial and final states, all the particles are out of each others' influence, however, in an intermediate time, these come together and *interact*. The amplitude for this process is given by:

$$\lim_{\tau\to\infty}\langle f|U(\tau,-\tau)|i\rangle\equiv\langle f|\mathcal{S}|i\rangle$$

where $\mathcal{S}$ is called the *scattering matrix* or the *S-matrix*. $\mathcal{S}$ is an unitary operator. In almost all cases, we will neglect the situation where no scattering has occurred, in which case we can define

$$\mathcal{S}=\mathbb{1}+i\mathcal{T}$$

and we will focus on the transfer matrix $\mathcal{T}$ henceforth. Till now we have been using the correlation function for all our interactions, however, it seems that the scattering amplitude is a more suitable quantity to measure in experimental observations.

There exists a thing called **Lehmann-Symanzik-Zimmermann (LSZ)** reduction formula which relates the correlation function with the scattering amplitude. The proof seems to be very complicated (for me), hence I am just going to mention the idea behind it.

Basically, for scattering amplitude, you are really interested in the actual *interaction* however, the correlation function also contains the free propagators.

What I mean is that, consider the diagram with the loop. For scattering processes, we generally know the initial and final momenta (momenta with which the particles are entering or going out), so we do not need to care about them. Thus, from the contribution, we remove the propagators of these 'known' momenta. This process is called *amputation*, where we remove the contribution from the external legs and this gives us the scattering matrix element.

There is always a delta function imposing momentum conservation which might become cumbersome to write everytime, hence we define the matrix $\mathcal{M}$ such that,

$$\langle\mathbf{p}_1\ldots\mathbf{p}_n|i\mathcal{T}|\mathbf{k}_1\ldots\mathbf{k}_m\rangle=(2\pi)^4\delta^{(4)}\left(\sum_i p_i-\sum_j k_j\right)(i\mathcal{M})$$

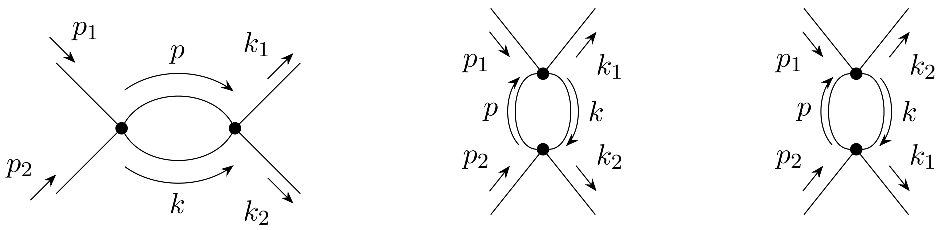
where p_i and k_j are the final and initial momenta. The matrix $\mathcal{M}$ will have contributions from the non-external propagators of all the diagrams that is possible for the process. In a gist, we have

$$\boxed{\mathcal{M}\equiv\text{ sum of all connected, amputated diagrams}}$$

For example, consider a process

$$p_1+p_2\longrightarrow k_1+k_2$$

Apart from the $\mathcal{O}(\lambda)$ diagram shown before, there are three order $\mathcal{O}(\lambda^2)$ diagrams which can represent the above scattering process and we have to consider all three of them:



- In the first diagram, momentum conservation gives $p_1 + p_2 = p + k = p_3 + p_4 \implies p = p_1 + p_2 - k$
- In the second diagram, momentum conservation gives $p_1 + p = k_1 + k \implies p = k_1 + k - p_1$
- In the first diagram, momentum conservation gives $p_1 + p = k_2 + k \implies p = k_2 + k - p_1$

The $\mathcal{O}(\lambda)$ process will contribute a factor of only $-i\lambda$ in the scattering matrix, since it has no loops (such a diagram is called a *tree-level diagram*) and hence no undetermined momenta to integrate out. Let us define the following quantity,

$$\mathcal{A}(p) \equiv \frac{(-i\lambda)^2}{2} \int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} \frac{i}{(p-k)^2 - m^2 + i\epsilon}$$

Using $\mathcal{A}(p)$, we can write the matrix $\mathcal{M}$ upto $\mathcal{O}(\lambda^2)$ for the scattering process $p_1 + p_2 \longrightarrow k_1 + k_2$:

$$i\mathcal{M} = \underbrace{(-i\lambda)}_{\text{tree-level}} + \mathcal{A}(p_1 + p_2) + \mathcal{A}(p_1 - k_1) + \mathcal{A}(p_1 - k_2)$$

Lecture 33: Scalar Yukawa Theory

We have dealt with ϕ^4 interaction in the previous few sections and we saw some Feynman diagrams for the interaction. Now we will consider the interaction between a real scalar and a complex scalar field, which goes by the name of *Yukawa interaction* which is of the form:

$$\mathcal{H} = g\psi^\dagger\psi\phi$$

where ψ is our charged complex field and ϕ is the real scalar field. The full Lagrangian is given by:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{1}{2}m^2\phi^2 + (\partial_\mu\psi^\dagger)(\partial^\mu\psi) - M^2\psi^\dagger\psi - g\psi^\dagger\psi\phi$$

where m is the mass associated with the ϕ field and M is the mass associated with the ψ field. The $1/2$ factor does not come with the complex scalar field, since a complex scalar field already has two degrees of freedom. And regarding the interaction, the simplest case is considered. $\psi^\dagger\psi$ gives a real, Lorentz invariant quantity and we multiply it with ϕ to get us an interaction between the two fields.

Having been familiarised with Feynman diagrams before, we will straightaway jump to calculate the amplitude of some scattering processes for this interaction. Before that we need to consider a few things:

- We have two kinds of field here, so we need a way to distinguish between them. We will denote the real scalar field by *dashed lines* while the complex scalar field by *solid lines*. Also, the solid lines will be *directed*, to denote the charge flow of the complex scalar field.

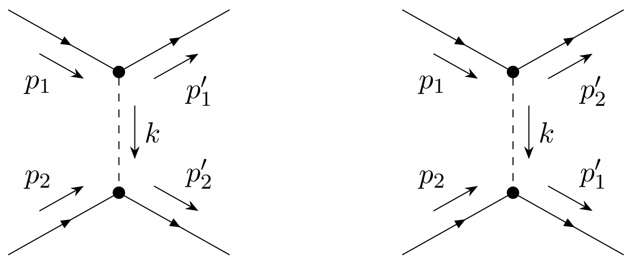
  The convention will be to have an *incoming* arrow for the initial state in ψ. For the initial state in $\psi^\dagger$, we will have an *outgoing* arrow. For the final state, the convention is reversed.

- Each interaction vertex will have *three* fields, since the interaction term has three fields. In ϕ^4 theory, we have tetravalent interaction vertices since the interaction had four terms in ϕ.

- The Feynman rules mostly remain same in this case. The propagator for real and complex field is the same, only the appropriate mass of the field needs to be taken in each case. Each vertex will have a factor of $(-ig)$.

Consider the process $\underline{\psi\psi \longrightarrow \psi\psi}$ where two incoming ψ particles result in two outgoing ψ particles. Henceforth, we will call ψ particles are *nucleons* and ϕ particles as *mesons*. Hence this is a case of nucleon-nucleon scattering.

At order $\mathcal{O}(g^0)$ we have no scattering, both ψ comes and leaves as it is, without any interaction. Diagram of order $\mathcal{O}(g)$ with only one vertex is not possible for this process. Hence, the only meaningful diagram will start at order $\mathcal{O}(g^2)$. The simplest diagrams contributing to this process are,



Explicitly written with the momenta, this scattering process reads

$$\psi(p_1) + \psi(p_2) \longrightarrow \psi(p_1') + \psi(p_2')$$

Consider the first diagram. Momentum conservation leads to $k = p_1 - p_1' = p_2' - p_2$. Since there are two vertices, the contribution of the vertex will be $(-ig)^2$. We will consider only the propagator for k, since all the other momenta are determined. Then the scattering amplitude for the first diagram will be,

$$i\mathcal{M}_1 = (-ig)^2 \frac{i}{(p_1 - p_1')^2 - m^2 + i\epsilon}$$

where we have substituted $k = p_1 - p_1'$ and since k is associated with the ϕ field, we considered the mass m. Similarly, for the second diagram we have,

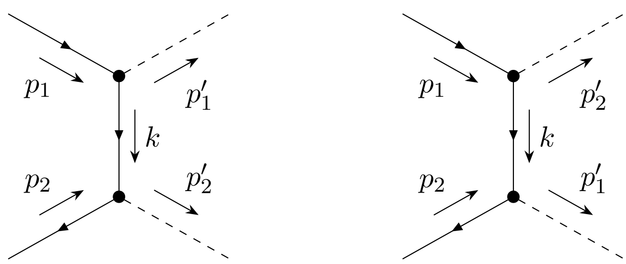
$$i\mathcal{M}_2 = (-ig)^2 \frac{i}{(p_1 - p_2')^2 - m^2 + i\epsilon}$$

The total scattering amplitude is the sum of these two terms,

$$i\mathcal{M} = (-ig)^2 \left[\frac{i}{(p_1 - p_1')^2 - m^2 + i\epsilon} + \frac{i}{(p_1 - p_2')^2 - m^2 + i\epsilon} \right]$$

These diagrams are tree-level diagrams and have no loops and hence no undetermined momentum to integrate over. The loops come in at $\mathcal{O}(g^4)$ and higher orders. Let us consider another process,

$$\psi(p_1) + \psi^\dagger(p_2) \longrightarrow \phi(p_1') + \phi(p_2')$$

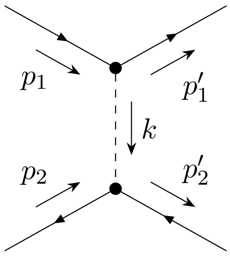


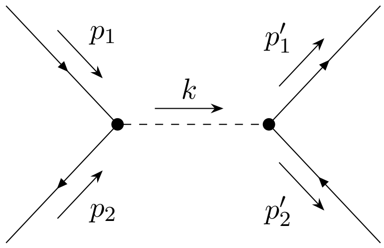
Observe the direction of the arrows denoting the charges. These have been drawn taking care of the *charge conservation* at each vertices. Since the internal line is now a *nucleon*, the amplitude is,

$$i\mathcal{M} = (-ig)^2\left[\frac{i}{(p_1-p_1')^2 - M^2 + i\epsilon} + \frac{i}{(p_1-p_2')^2 - M^2 + i\epsilon}\right]$$

Consider yet another process, which will have a bit different diagram than the previous two

$$\psi(p_1) + \psi^\dagger(p_2) \longrightarrow \psi(p_1') + \psi^\dagger(p_2')$$





The second diagram is a tad different from the other diagrams that we have seen. Momentum conservation in the second diagram gives $p_1 + p_2 = k$. Thus the amplitude for the process will be given by,

$$i\mathcal{M} = (-ig)^2\left[\frac{i}{(p_1-p_1')^2 - m^2 + i\epsilon} + \frac{i}{(p_1+p_2)^2 - m^2 + i\epsilon}\right]$$

2.1 Mandelstam variables

The quantities $(p_1 - p_1')$, $(p_1 - p_2')$ and $(p_1 + p_2)$ seems to appear in the propagator expression quite often. These are given some standard names, called *Mandelstam variables*, which are defined as the Lorentz invariant quantities,

$$\begin{aligned} s &= (p_1+p_2)^2 = (p_1'+p_2')^2 \\ t &= (p_1-p_1')^2 = (p_2-p_2')^2 \\ u &= (p_1-p_2')^2 = (p_2-p_1')^2 \end{aligned}$$

These are also called s, t, u-channels. Firstly note that we have the total momentum conservation, $p_1 + p_2 = p_1' + p_2'$. Then,

$$\begin{aligned} s+t+u &= (p_1+p_2)^2 + (p_1-p_1')^2 + (p_1-p_2')^2 \\ &= p_1^2 + p_2^2 + 2p_1\cdot p_2 + p_1^2 + p_1'^2 - 2p_1\cdot p_1' + p_1^2 + p_2'^2 - 2p_1\cdot p_2' \\ &= 3p_1^2 + p_2'^2 + p_1'^2 + p_2^2 + 2p_1\cdot(p_2 - p_1' - p_2') \\ &= 3p_1^2 + p_2'^2 + p_1'^2 + p_2^2 - 2p_1^2 \\ &= p_1^2 + p_2'^2 + p_1'^2 + p_2^2 \end{aligned}$$

where $p_i^2 \equiv (p_i)^\mu (p_i)_\mu$ and $p_i \cdot p_j \equiv (p_i)^\mu (p_j)_\mu$ are the Lorentz scalar product. Using the on-shell condition, $p_j^2 = m_j^2$ we have $s + t + u = \sum m_j^2$, which is a nice thing perhaps. Now, consider the *centre of mass/momentum* frame, where the total three-momentum is zero. Then,

$$\begin{aligned} \mathbf{p}_1 + \mathbf{p}_2 = 0 &\implies \mathbf{p}_i \equiv \mathbf{p}_1 = -\mathbf{p}_2 \\ \mathbf{p}_1' + \mathbf{p}_2' = 0 &\implies \mathbf{p}_f \equiv \mathbf{p}_1' = -\mathbf{p}_2' \end{aligned}$$

From the above conditions, the momentum four vectors become

$$p_1 = (E_1, \mathbf{p}_i), p_2 = (E_2, -\mathbf{p}_i), p_1' = (E_1', \mathbf{p}_f), p_2' = (E_2', -\mathbf{p}_f)$$

Then, $s = (p_1+p_2)^2 \equiv (E_1+E_2)^2 \equiv E_{\mathrm{T}}^2$. s thus measures the total center of mass energy of the collision.

We can now find an expression for the energies in terms of the variable s. For that, we will require the on-shell condition, $p^2 = m^2$ and the above relation, which gives us:

$$E_1 + E_2 = \sqrt{s} \qquad E_1^2 - \mathbf{p}_i^2 = m_1^2 \qquad E_2^2 - \mathbf{p}_i^2 = m_2^2$$

Using the above, we get the following,

$$m_1^2 - m_2^2 = E_1^2 - E_2^2 = E_1^2 - (\sqrt{s} - E_1)^2 = \sqrt{s}(2E_1 - \sqrt{s}) \implies E_1 = \frac{s + m_1^2 - m_2^2}{2\sqrt{s}}$$

Similarly, we obtain the other energies in terms of the masses and s,

$$E_2 = \frac{s + m_2^2 - m_1^2}{2\sqrt{s}} \qquad E_1' = \frac{s + m_1'^2 - m_2'^2}{2\sqrt{s}} \qquad E_2' = \frac{s + m_2'^2 - m_1'^2}{2\sqrt{s}}$$

Let us now consider the variable $t = (p_1 - p_1')^2$ which turns out to be

$$\begin{aligned} t &= p_1^2 + p_1'^2 - 2p_1 \cdot p_1' \\ &= p_1^2 + p_1'^2 - 2((p_1)^0(p_1')^0 - |\mathbf{p}_1||\mathbf{p}_1'|\cos\theta) \\ &= p_1^2 + p_1'^2 - 2(E_1E_1' - |\mathbf{p}_1||\mathbf{p}_1'|\cos\theta) \end{aligned}$$

where θ is the angle between $\mathbf{p}_1$ and $\mathbf{p}_1'$. Let us take the elastic limit, $m_1 = m_1'$ and $m_2 = m_2'$ which gives us $E_1 = E_1' = \mathfrak{E}_1$ and $E_2 = E_2' = \mathfrak{E}_2$ along with $|\mathbf{p}_i| = |\mathbf{p}_f| \equiv \mathfrak{p}$. Then the expression for t becomes,

$$t = \cancel{2\mathfrak{E}_1^2} - 2\mathfrak{p}^2 - \cancel{2\mathfrak{E}_1^2} + 2\mathfrak{p}^2\cos\theta = -2\mathfrak{p}^2(1 - \cos\theta) = -4\mathfrak{p}^2\sin^2\left(\frac{\theta}{2}\right) \le 0$$

In a similar way, we can derive $u = -4\mathfrak{p}^2\cos^2\left(\frac{\theta}{2}\right) \le 0$. Thus, t and u are measures of the momentum exchanged between particles. The amplitudes of the scattering processes can be written entirely in terms of the Mandelstam variables.

- For the process $\psi\psi \longrightarrow \psi\psi$ we have $i\mathcal{M} = (-ig)^2\left[\frac{i}{t-m^2+i\epsilon} + \frac{i}{u-m^2+i\epsilon}\right]$
- For the process $\psi\psi^\dagger \longrightarrow \phi\phi$ we have $i\mathcal{M} = (-ig)^2\left[\frac{i}{t-M^2+i\epsilon} + \frac{i}{u-M^2+i\epsilon}\right]$
- For the process $\psi\psi^\dagger \longrightarrow \psi\psi^\dagger$ we have $i\mathcal{M} = (-ig)^2\left[\frac{i}{t-m^2+i\epsilon} + \frac{i}{s-m^2+i\epsilon}\right]$

Lecture 33: Spinor Yukawa Theory